OSHW_3_teambased_112590013_112590019_112590023_112590028

Chap7,8,9

Compilation and Configuration Instructions

Change directory to the folder, then run the following commands in the terminal:

Chap7: gcc thread_pool.c -o thread_pool -lpthread

Chap8: gcc banker.c -o banker -lpthread

Chap9: gcc memory.c -o memory

Then run the executable files

Execution Snapshots:

Chap 7

```
zxzy2@MSI:/mnt/c/Users/zxzy2/OneDrive/Desktop/
Thread 124755910846144 working on task 1
Thread 124755919238848 working on task 2
Thread 124755894060736 working on task 0
Thread 124755902453440 working on task 3
Thread 124755910846144 working on task 4
Thread 124755894060736 working on task 5
Thread 124755919238848 working on task 6
Thread 124755902453440 working on task 7
Thread 124755910846144 working on task 8
Thread 124755894060736 working on task 9
Thread 124755919238848 working on task 10
Thread 124755902453440 working on task 11
Thread 124755910846144 working on task 12
Thread 124755919238848 working on task 13
Thread 124755894060736 working on task 14
Thread 124755902453440 working on task 15
Thread 124755910846144 working on task 16
Thread 124755894060736 working on task 17
Thread 124755919238848 working on task 18
Thread 124755902453440 working on task 19
Thread 124755910846144 working on task 10
Thread 124755902453440 working on task 11
Thread 124755894060736 working on task 12
Thread 124755919238848 working on task 13
zxzy2@MSI:/mnt/c/Users/zxzy2/OneDrive/Desktop/
```

Chap 8

```
zxzy2@MSI:/mnt/c/Users/zxzy2/OneDrive/Desktop/code/NTUT/OperatingSystem/HW3/Team/Chap8$ ./banker 4 2 3
Initializing system...

===== SYSTEM STATE =====
Available: 4 2 3
Customer 1 requesting: 0 1 2
→ GRANTED

===== SYSTEM STATE =====
Available: 4 1 1
Customer 3 requesting: 0 0 1
→ GRANTED

===== SYSTEM STATE =====
Available: 4 1 0
Customer 2 requesting: 0 1 1
→ Denied (not enough resources)
Customer 0 requesting: 1 0 2
→ Denied (not enough resources)
Customer 4 requesting: 3 2 2
→ Denied (not enough resources)
Customer 4 requesting: 0 0 0
→ GRANTED
```

```
===== SYSTEM STATE =====
Available: 4 1 0
Customer 2 requesting: 1 1 1
→ Denied (not enough resources)
Customer 3 releasing: 0 0 1

===== SYSTEM STATE =====
Available: 4 1 1
Customer 1 releasing: 0 1 2

===== SYSTEM STATE =====
Available: 4 2 3
Customer 0 requesting: 1 0 2
→ GRANTED

===== SYSTEM STATE =====
Available: 3 2 1
Customer 4 releasing: 0 0 0
```

Chap 9

```
zxzy2@MSI:/mnt/c/User
Allocated at 0
Allocated at 100
Released 0
[0-99] Free
[100-299] Allocated
[300-1023] Free
zxzy2@MSI:/mnt/c/User
```

Team members and responsibilities:

Student_ID	Name	Responsibility
112590013	陳岱佑	Ch08. Proj, Documentation
112590019	謝政軒	Ch08. Proj, Snapshots
112590023	林宇祥	Ch07. Proj
112590028	高鉉	Ch09. Proj